

# Pierre Hollebèque

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As a second-year Master's student in Artificial Intelligence and Machine Learning at DTU, I am looking for a student position where I can leverage my skills in deep learning to contribute to innovative, real-world projects.

## Education

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| <b>Technical University of Denmark, MSc Human-Centered Artificial Intelligence</b> | Sept 2025 - July 2027 |
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- Double-degree with Centrale Lyon.
- **AI & Machine Learning:** Deep Learning, Advanced Machine Learning, Graph Neural Networks, Bayesian Machine Learning, Multi-Agent Systems.
- **Data Engineering & Computing:** Computational Tools for Data Science, Algorithms for Massive Datasets, Python HPC, Social Graphs.
- **Human-Computer Interaction:** UX Design.

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| <b>Centrale Lyon, MSc in General Engineering</b> | Sept 2023 – July 2027 |
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GPA: 3.92/4 - Completed Core Engineering Program with Highest Honors

- **General Courses:** Mathematics, Physics, Computer Science, Telecommunication, Electronics, Mechanical Engineering, and Economics.
- **Specialization:** Probabilities & Statistics, Genetic Algorithms & Ant Colony Optimization (ACO), Image Sensing and Processing, Machine Learning, Constraint Logic Programming.

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| <b>Lycée Chateaubriand, Preparatory classes (MPSI-MP*)</b> | Sept 2021 – July 2023 |
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- Two-year intensive French program preparing for the competitive entrance exams to top engineering schools.
- Curriculum focused on Advanced Mathematics, Physics, Computer Science, and theoretical rigor.

## Experience

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| <b>Generative AI Intern, NIJI – Rennes, France</b> | Apr 2025 – Aug 2025 |
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- **Design of an AI-powered Learning Solution:** Led the end-to-end conception of a corporate training tool designed to optimize memory retention and skill acquisition.
- **Neuroscience-based:** Conducted state-of-the-art research on cognitive learning methods to integrate evidence-based educational principles into product features.
- **RAG Implementation:** Leveraged Retrieval-Augmented Generation (RAG) to transform static training repositories into an interactive and searchable knowledge base.
- **Agentic Workflow Automation:** Designed AI agent workflows to automatically generate question banks and assessment materials from existing documentation.
- **Adaptive Learning Algorithm:** Conceptualized a recommendation engine that personalizes learning paths by adjusting question difficulty to the learner's proficiency.

## Projects

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| <b>Diffusion-based sampling for approximate posterior inference, DTU</b> | Sep 2026 - now |
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- **Bayesian Machine Learning Project Course** throughout the fall semester.

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| <b>Multi-Agent Path Finding (MAPF) &amp; Cooperative RL, DTU</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | Feb 2026 - May 2026 |
| <ul style="list-style-type: none"> <li>• <b>Classical Search (MAPF):</b> Implemented A*, BFS, DFS, and Enhanced Conflict-Based Search (EECBS) to compute optimal and bounded-suboptimal plans in a grid-based Hospital domain.</li> <li>• <b>Deep Reinforcement Learning (DRL):</b> Trained a decentralized agent using Proximal Policy Optimization (PPO) to learn cooperative policies for a multi-agent box-pushing environment.</li> </ul>                                             |                     |
| <b>Feature alignment for unpaired infrared image translation in the microstructures of composite materials, DTU</b>                                                                                                                                                                                                                                                                                                                                                                        | Nov 2025 - Dec 2025 |
| <ul style="list-style-type: none"> <li>• <b>Infrared Super-Resolution:</b> Developed a Cycle-GAN solution (Python, PyTorch) trained on unpaired data to upscale low-resolution sensor outputs into high-fidelity images, reducing hardware costs for experiment monitoring.</li> </ul>                                                                                                                                                                                                     |                     |
| <b>Analysis of the Evolution of Alliances and Polarization in the French National Assembly, DTU</b>                                                                                                                                                                                                                                                                                                                                                                                        | Nov 2025 - Dec 2025 |
| <ul style="list-style-type: none"> <li>• <b>Graph Theory &amp; Network Analysis:</b> Constructed legislative graphs based on roll-call votes to model political alliances.</li> <li>• <b>Unsupervised Learning:</b> Applied Louvain community detection to identify evolving political blocs and quantify polarization.</li> <li>• <b>NLP &amp; Text Mining:</b> Implemented an adapted TF-IDF approach to characterize the linguistic priorities of each identified community.</li> </ul> |                     |
| <b>Research Project - State-of-the-art review of Transformer architecture in AI, Liris Laboratory &amp; Centrale Lyon</b>                                                                                                                                                                                                                                                                                                                                                                  | Oct 2024 - Apr 2025 |
| <ul style="list-style-type: none"> <li>• <b>Theoretical Research:</b> Conducted a comprehensive state-of-the-art review of Transformer models and authored a technical report on ethical challenges.</li> <li>• <b>Practical Implementation:</b> Built a Seq2Seq Transformer model from scratch, fine-tuned a BERT model, and developed a RAG system using Mistral-7B.</li> </ul>                                                                                                          |                     |
| <b>Study Project - Modeling wildfire behavior and crowd dynamics, Centrale Lyon</b>                                                                                                                                                                                                                                                                                                                                                                                                        | Oct 2023 - May 2024 |
| <ul style="list-style-type: none"> <li>• <b>Simulation &amp; Leadership:</b> Led a team of 6 to develop a computational simulation model for wildfire behavior and crowd dynamics. Awarded the <b>Francis Leboeuf Prize</b> for Best Project (2023).</li> </ul>                                                                                                                                                                                                                            |                     |

## Skills

**Programming Languages:** Python, Matlab, SQL, Prolog.

**Deep Learning & Computer Vision:** PyTorch, TensorFlow, OpenCV, NumPy, SciPy, Scikit-learn, Hugging Face.

**Tools & Platforms:** Git, GitHub, Streamlit, Figma, HPC (Remote Supercomputing).

**Languages:** French (Native), English (C1).